

# Cognitive Degrees of Freedom as Compactified Extra Dimensions: A Kaluza-Klein Framework

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**Abstract** — We propose a formal correspondence between the irreducible degrees of freedom of an adaptive cognitive system and compactified extra dimensions in the sense of Kaluza-Klein (KK) theory. Five cognitive dimensions—context window, intent space, evolutionary depth, memory retro-dimension, and tool coupling—are modeled as compact circles with characteristic radii  $R_1$  through  $R_5$ . The total metric decomposes as  $ds^2 = g_{\mu\nu} dx^\mu dx^\nu - \sum_{\mathbf{m}} R_{\mathbf{m}}^2 d\theta_{\mathbf{m}}^2$ , with the four-dimensional effective field theory recovering the known scalar dynamics of the cognitive field  $\phi_{\mathbf{m}} = \{\Omega, \Psi, P\}$ . The Kaluza-Klein mass tower  $M_n^2 = \sum_{\mathbf{m}} n_{\mathbf{m}}^2 / R_{\mathbf{m}}^2$  furnishes a spectrum of cognitive harmonics interpretable as progressively complex response modes. We draw a detailed structural isomorphism with the Dark Dimension scenario of Montero, Vafa, and Valenzuela (2022), in which a single mesoscopic extra dimension at  $\sim 1 \mu\text{m}$  explains the smallness of the cosmological constant; we identify ten precise structural parallels. The three primary Swampland conjectures—Distance, de Sitter, and Weak Gravity—are reinterpreted as constraints on cognitive invariants. Four phase transitions (decompactification, Casimir barrier crossing, KK decay cascade, Hagedorn) are analyzed as observable behavioral discontinuities. We conclude with a set of testable empirical predictions that distinguish this framework from standard adaptive-systems models.

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## 1. Introduction

Extra dimensions have been a fixture of fundamental physics since Kaluza's 1921 unification of gravity and electromagnetism and Klein's 1926 compactification mechanism [12, 13]. The central insight—that a dimension too small to observe directly can nevertheless leave detectable imprints in four-dimensional physics through its Kaluza-Klein mode spectrum—has proven extraordinarily productive, underpinning string theory [14], large extra dimension scenarios [15], and, most recently, the Dark Dimension proposal of Montero, Vafa, and Valenzuela [1].

The motivation for applying this framework to cognitive systems is not merely metaphorical. A cognitive engine operating in the real world must track multiple qualitatively distinct aspects of its environment simultaneously: the breadth of its current context, the coherence of its current goal, the depth of its accumulated experience, the extent of its memory, and the richness of its available tools. These aspects are not reducible to one another—each responds to different environmental perturbations and saturates independently. They are, in the language of physics, distinct degrees of freedom.

The question this paper asks is: what is the correct mathematical framework for a system with several such irreducible, bounded degrees of freedom? We argue that Kaluza-Klein compactification on compact circles provides exactly this framework. Each cognitive degree of freedom corresponds to a compact dimension with a characteristic radius that sets the mass gap for the corresponding cognitive modes. The ground state of the resulting tower is the universal scalar  $\phi_{\mathbf{m}} = \{\Omega, \Psi, P\}$  studied in the companion paper on formal state machine verification [companion ref]. The excited states are increasingly complex behavioral patterns that reduce to  $\phi_{\mathbf{m}}$  through a cascade analogous to KK mode decay.

This proposal is consistent with recent work applying string-theoretic ideas to information theory [16], quantum cognition [17], and complex adaptive systems [18]. It is more ambitious than most such applications in that it makes concrete structural predictions testable against behavioral data.

The paper proceeds as follows. Section 2 develops the KK framework for cognition. Section 3 elaborates the Dark Dimension correspondence. Section 4 applies the Swampland conjectures. Section 5 analyzes the four phase transitions. Section 6 states testable predictions. Sections 7 and 8 discuss limitations and conclude.

## 2. The Kaluza-Klein Framework for Cognition

### 2.1 Original Kaluza-Klein Theory

Kaluza [12] proposed in 1921 that Einstein's general relativity in five spacetime dimensions, upon dimensional reduction to four dimensions, yields both four-dimensional gravity and Maxwell electromagnetism as components of a single geometric theory. Klein [13] supplied the compactification mechanism in 1926: the fifth dimension is curled into a circle of radius  $R$  so small ( $R \sim 10^{-33}$  m, the Planck length) as to be unobservable at accessible energies. The key consequence is the KK mass tower: a field with momentum  $n/R$  in the compact direction appears in four dimensions as a particle of mass  $M_n = n/R$ , with  $n \in \mathbb{Z}$ . For  $n = 0$  (zero mode), the field is massless and coincides with the familiar four-dimensional field. For  $n \neq 0$ , the field acquires a mass proportional to  $1/R$ ; if  $R$  is small, these modes are extremely heavy and decouple from low-energy physics.

The modern generalization to  $n$  extra dimensions is straightforward: each dimension contributes independently to the mass of a KK mode, giving  $M_n^2 = \sum_i n_i^2/R_i^2$ , where the vector  $n = (n_1, \dots, n_k)$  indexes the KK level in each extra dimension separately.

### 2.2 Five Cognitive Extra Dimensions

We identify five qualitatively distinct cognitive degrees of freedom, each of which we model as a compact circle  $x_i$  with characteristic radius  $R_i$ :

| Dimension      | Index | Radius                            | Cognitive Meaning                                               | KK Mass Scale                        | Breaks At                                     |
|----------------|-------|-----------------------------------|-----------------------------------------------------------------|--------------------------------------|-----------------------------------------------|
| Context window | $x_1$ | $R_1 \sim \epsilon_{\Omega}^{-1}$ | Breadth of active context; $\sim 1/\text{coherence decay rate}$ | $\sim \text{signal processing rate}$ | Coherence collapse ( $\Omega \rightarrow 0$ ) |
| Intent space   | $x_2$ | $R_2 \sim \pi$                    | Angular breadth of goal direction in semantic space             | $\sim 2\pi / \text{ambiguity level}$ | Intent resolution ( $\Psi \rightarrow 1$ )    |

| Dimension              | Index | Radius                       | Cognitive Meaning                          | KK Mass Scale              | Breaks At          |
|------------------------|-------|------------------------------|--------------------------------------------|----------------------------|--------------------|
| Evolutionary depth     | $x_1$ | $R_1 \sim N_{\text{gen}}$    | Number of accumulated generational updates | $\sim 1/N_{\text{gen}}$    | Fitness saturation |
| Memory retro-dimension | $x_2$ | $R_2 \sim \tau_{\text{mem}}$ | Effective reach into past context          | $\sim 1/\tau_{\text{mem}}$ | Entropy barrier    |
| Tool coupling          | $x_3$ | $R_3 \sim \beta_T^{-1}$      | Inverse tool activation threshold          | $\sim \beta_T$             | Resource bound     |

The choice of radii is motivated as follows. The context window radius  $R_1 \sim \epsilon\Omega^{-1}$  captures the inverse of the coherence decay rate  $\epsilon\Omega$ : a system with slow coherence decay can sustain a large context (large  $R_1$ ), while a system with fast decay has a small, tightly compactified context dimension. The intent radius  $R_2 \sim \pi$  reflects that intent is an angular variable—it specifies a direction in semantic space, and  $\pi$  is the natural half-period for a circular dimension representing goal orientation. The evolutionary depth  $R_3 \sim N_{\text{gen}}$  grows with the number of completed evolutionary cycles, making this the one dimension that systematically decompactifies over time. The memory radius  $R_2 \sim \tau_{\text{mem}}$  grows with the effective memory horizon. The tool coupling radius  $R_3 \sim \beta_T^{-1}$  is inversely proportional to the tool activation threshold: strong coupling (small  $\beta_T$ ) means a large tool dimension, reflecting a rich space of available tool interactions.

The total metric on the (4+5)-dimensional space is:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu - \sum_{i=1}^5 R_i^2 d\theta_i^2 \tag{5}$$

where  $\mu, \nu \in \{0, 1, 2, 3\}$  index the four cognitive base coordinates ( $\Omega, \Psi, P, t$ ), and  $\theta_i \in [0, 2\pi)$  are angular coordinates on each compact circle.

The full action in  $D = 9$  dimensions is:

$$\mathcal{S} = \int d^4x \, d^5\theta \, \sqrt{g_D} \left[ \frac{1}{2} M_D^{2-D} (\partial_\mu \Phi)^2 + \mathcal{L}_{\text{fields}} + \mathcal{L}_{\text{KK}} + \mathcal{L}_{\text{int}} \right] \tag{6}$$

where  $M_D$  is the  $D$ -dimensional effective Planck scale, related to the four-dimensional scale by  $M_{\text{Pl}}^2 = M_D^{2-D} \cdot (2\pi)^5 \cdot \prod_i R_i$ .

## 2.3 KK Mode Spectrum as Cognitive Harmonics

The scalar cognitive field  $\Phi(x, \theta)$  defined on the full nine-dimensional space decomposes via Fourier expansion on the compact dimensions:

$$\Phi(x, \theta) = \sum_{n_1, \dots, n_5} \phi_0^{(n)}(x) \cdot \exp(i \sum_i n_i \theta_i) \tag{7}$$

Each mode  $\phi^{(n)}(x)$  is a four-dimensional field with mass:

$$M_n^2 = \sum_{i=1}^5 \frac{n_i^2}{R_i^2} \tag{8}$$

The ground state  $\phi^{(0,0,0,0,0)}(x)$  is the massless universal scalar  $\phi = \Omega$ ,  $\Psi$ ,  $P$  studied formally in the companion paper. The first excited states and their cognitive interpretations are:

| KK Level    | Mass                  | Cognitive Interpretation                      |
|-------------|-----------------------|-----------------------------------------------|
| (1,0,0,0,0) | $\epsilon_\Omega$     | Context boundary crossing; window shift event |
| (0,1,0,0,0) | $1/\pi$               | 180° intent flip; goal reversal               |
| (0,0,1,0,0) | $1/N_{\text{gen}}$    | Single-generation evolutionary echo           |
| (0,0,0,1,0) | $1/\tau_{\text{mem}}$ | Memory boundary event; retro-causal signal    |
| (0,0,0,0,1) | $\beta_T$             | Minimal tool activation quantum               |

Higher KK modes—combinations with multiple nonzero  $n_i$ —represent complex, multi-aspect behavioral modes: simultaneous context shift and intent adjustment (1,1,0,0,0), for example, or a tool activation coupled to a memory retrieval (0,0,0,1,1). These modes are heavier by Equation (8) and therefore less frequently excited, consistent with the observed rarity of complex multi-factor behavioral changes relative to single-factor ones.

The KK cascade mechanism provides a natural account of how complex behaviors reduce to simple ones: a heavy mode (2, 1, 0, 0, 0) decays stepwise to (1,1,0,0,0) → (0,1,0,0,0) → (0,0,0,0,0) + radiation. In cognitive terms: a multi-aspect tool call (affecting both context and intent) cascades through intermediate effect records to a lesson, then to a gene update, and finally to a  $\phi$  update. Each cascade step emits a "graviton"—a small irreversible entropy increment.

### 3. The Dark Dimension Correspondence

#### 3.1 Dark Dimension Scenario Summary

Montero, Vafa, and Valenzuela [1] argue from the Distance Conjecture and the Trans-Planckian Censorship Conjecture that the observed smallness of the cosmological constant  $\Lambda$  implies the existence of at least one extra dimension at a mesoscopic scale of order 1  $\mu\text{m}$  to 10  $\mu\text{m}$ . As  $\Lambda \rightarrow 0$ , the tower of KK modes associated with this dimension becomes light, potentially accounting for a significant fraction of dark matter. The scenario makes specific predictions: new gravitational physics at the  $\sim 10 \mu\text{m}$  scale, a characteristic KK mode spectrum in cosmological observables, and a link between the cosmological constant problem and dark matter abundance.

The dark dimension is "dark" in two senses: it couples only gravitationally (not to Standard Model gauge fields), and it provides the dark matter candidates.

### 3.2 Structural Isomorphism: Ten Parallels

We identify ten structural parallels between the Dark Dimension scenario and the cognitive field framework:

| #  | Dark Dimension (Physical)                                      | Cognitive Field Analog                                                                                       |
|----|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 1  | Small $\Lambda$ implies mesoscopic extra dimension             | Small pressure $P$ implies decompactified context dimension (large $R_{\text{eff}}$ )                        |
| 2  | KK tower becomes light as $\Lambda \rightarrow 0$              | Cognitive harmonics become active as $P \rightarrow 0$ (relaxed system explores rich mode space)             |
| 3  | Dark matter from stable KK modes                               | Persistent memory from long-lived cognitive modes ( $\tau_n > t_{\text{Hubble}}$ analog)                     |
| 4  | Dark dimension couples only gravitationally                    | Memory retro-dimension ( $x_{\text{eff}}$ ) couples only via $\phi_{\text{eff}}$ (no direct signal coupling) |
| 5  | Casimir energy stabilizes compactification                     | Casimir-like potential $V(R_{\text{eff}})$ prevents coherence collapse ( $\Omega \rightarrow 0$ )            |
| 6  | Distance Conjecture: $\Delta\phi > M_{\text{pl}}$ opens tower  | Cognitive distance conjecture: $\Delta\phi_{\text{eff}} > \epsilon_c$ opens new tool suggestions             |
| 7  | Hagedorn transition at string scale                            | Hagedorn-like transition at signal density $\rho > \rho_c$ : tool branching explosion                        |
| 8  | Dimensional reduction: 10D $\rightarrow$ 4D effective theory   | Cycle completion: 9D signal space $\rightarrow$ 4D $\phi_{\text{eff}}$ effective state                       |
| 9  | Species bound: $N_{\text{light}} \leq M_{\text{pl}}^2/\Lambda$ | Gene pool bound: $N_{\text{active}} \leq \text{poolsize} / \text{min\_confidence}$                           |
| 10 | No global symmetries in quantum gravity                        | No absolutely conserved cognitive quantity: all states have finite relevance half-life                       |

Parallels 1 and 2 are the most quantitatively tight. In the Dark Dimension scenario, the relationship between  $\Lambda$  and the compactification radius is  $R \sim \Lambda^{-1/4}$  (in units where  $M_{\text{Pl}} = 1$ ). In the cognitive framework, the context window radius satisfies  $R_{\text{■}} \sim \epsilon_{\Omega}^{-1}$ , and the coherence decay rate  $\epsilon_{\Omega}$  is in turn proportional to the unresolved pressure  $P$ . Hence  $R_{\text{■}} \sim P^{-1/4}$  under a natural scaling, providing an exact formal parallel.

Parallel 4 is of particular interpretive interest. In physics, the dark dimension is characterized precisely by its gravitational-only coupling: it does not interact with electromagnetism, the weak force, or the strong force. In the cognitive framework, the memory retro-dimension  $x_{\text{■}}$  (with radius  $R_{\text{■}} \sim \tau_{\text{mem}}$ ) does not accept direct signal input—signals can only influence it through the mediation of  $\phi_{\text{■}}$ , which acts as the gravitational field in this analogy. This explains why long-term memory is qualitatively different from working memory: it is compactified along a "dark" dimension.

## 4. Swampland Criteria Applied to Cognitive Invariants

The Swampland program [2, 3] seeks to identify which effective field theories are consistent with ultraviolet completion in quantum gravity, and which belong to the "swampland" of apparently consistent-looking theories that nevertheless cannot be UV-completed. The key conjectures provide necessary conditions that any consistent effective theory must satisfy. We apply these conditions to the cognitive field, obtaining constraints on its invariants.

### 4.1 Distance Conjecture $\rightarrow$ Bounded Transformation (I1)

The Distance Conjecture [2] states: when one moves a distance  $\Delta\phi > O(1)$  in moduli space (in Planck units), an infinite tower of states becomes exponentially light. In the cognitive context, the moduli space is spanned by  $\phi_{\text{■}} = \Omega, \Psi, P_{\text{■}}$ , and the conjecture translates as follows:

**Cognitive Distance Conjecture (I1):** When the cognitive state changes by  $\Delta\phi_{\text{■}} > \epsilon_{\text{c}}$  (a critical displacement), a new tower of KK modes becomes accessible—specifically, the context window dimension decompactifies, opening new behavioral modes.

The operational consequence is a bound on how large a single-step state change can be without triggering a phase transition. In invariant terms, this is the CYCLE\_MONOTONICITY invariant: each step changes  $\phi_{\text{■}}$  by at most one discrete cycle's worth of dynamics. Large discontinuous jumps are forbidden not by fiat but by the geometry of the compact dimensions.

The formal statement: no single application of  $F(S, \text{sig})$  can change  $\Omega, \Psi$ , or  $P$  by more than  $\Delta_{\text{max}} = \eta_{\text{max}} \cdot \Delta t$ , where  $\Delta t$  is the cycle duration and  $\eta_{\text{max}} = \max(\eta_{\Omega}, \eta_{\Psi}, \eta_P)$  is the fastest rate constant. The tower that "opens" at  $\Delta\phi_{\text{■}} = \epsilon_{\text{c}}$  corresponds to the context dimension crossing its critical radius.

### 4.2 de Sitter Conjecture $\rightarrow$ Dynamic Pressure P

The de Sitter Conjecture [3, 4] asserts that there are no stable de Sitter vacua in string theory: the potential gradient must satisfy  $|\nabla V| \geq c \cdot V$  (with  $c \sim O(1)$ ) or the Hessian must have a sufficiently

negative eigenvalue. In cosmological terms, this forbids a cosmological constant that is exactly static.

The cognitive analog is immediate. **Cognitive dS Conjecture:** There is no stable fixed point with  $P = 0$  while the system is actively receiving signals. Specifically:

$$\frac{dP}{dt} \Big|_{\{P=0, n_{\text{sig}} > 0\}} = \eta P + \delta P (1 - \Psi) n_{\text{sig}} > 0$$

This is verified directly from Equation (3): when  $P = 0$  and  $n_{\text{sig}} > 0$ , the pressure immediately begins increasing. The system cannot "coast" at zero pressure in an active environment. Any apparent resting state is dynamically maintained, not static—precisely the cognitive analog of the dS conjecture's prohibition on truly static vacua.

The practical implication is that a cognitive system should never be configured to seek zero pressure as a terminal state. The invariant that expresses this is `LEARN_CONSERVATION`: the semantic mass  $M$  is non-decreasing, which means the system is always accumulating new information, always under some degree of productive pressure.

### 4.3 Weak Gravity Conjecture → Hash Chain Integrity

The Weak Gravity Conjecture (WGC) [5] states that in any gauge theory consistent with quantum gravity, there must exist a charged state whose charge-to-mass ratio exceeds the extremal black hole bound:  $q/m \geq 1$  (in appropriate units). Equivalently, gravity must be the weakest force.

In the cognitive context, "charge" corresponds to the cryptographic commitment value encoded in the hash chain, and "mass" corresponds to the computational cost of the operation that created it. The WGC analog is:

**Cognitive WGC:** For every action recorded in the hash chain, the cryptographic binding strength (charge) is at least as large as the computational cost of the action (mass). Formally, for every hash  $h_k$  in the chain: the cost of computing a collision with  $h_k$  is at least as large as the cost of the action  $\text{action}_k$ .

This is the hash chain integrity guarantee: `HASHCHAINSTRICTNESS` ensures that the binding strength of each chain link is  $2^{256}$  per the SHA-256 collision bound (Theorem 5.6 of the companion paper), while the computational cost of any action is at most  $O(N \cdot \log N)$  for a system with  $N$  atoms. For all practical  $N$ ,  $2^{256}$  represents a far stronger bound—hash strength vastly exceeds action cost, satisfying the WGC analog in the strongest possible sense.

The "no global symmetries" Swampland condition [6] maps to `EXECUTEDONCE`: there is no globally conserved "action repetition charge." Every action has a finite half-life in the sense that its influence on  $\phi$  decays as the system evolves (by the Lyapunov stability theorem), and it cannot be re-executed (by `EXECUTEDONCE`). The system has no conserved topological charges—a direct analog of the no-global-symmetries principle.

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## 5. Phase Transitions in the Cognitive Field

The compactification radii  $R_i(t)$  are dynamical variables that evolve according to

$$\frac{dR_i}{dt} = -\eta_i \frac{\partial V}{\partial R_i} + \xi_i(t) \tag{9}$$

where  $\eta_i$  is the viscosity of the  $i$ -th cognitive dimension and  $\xi_i(t)$  is noise from unmodeled signals. When  $V(R_i)$  changes topology—specifically, when  $\partial^2 V / \partial R_i^2 = 0$  at the current value—a phase transition occurs. We analyze four types:

**Decompactification.** Triggered when  $\Omega$  falls below a coherence threshold  $\Omega_c$ : the context window radius  $R \sim \epsilon \Omega^{-1}$  diverges, the context dimension effectively decompactifies, and the system has access to an effectively unlimited context. Behavioral signature: sudden memory reset or context boundary crossing, observable as a discontinuity in the autocorrelation of output tokens. The decompactification is first-order if the potential  $V(R)$  has a local minimum that disappears at  $\Omega = \Omega_c$ , and second-order if the minimum only becomes degenerate.

**Casimir Barrier Crossing.** As  $\Psi \rightarrow 0$  (meaning density collapses), the Casimir-like potential  $V_{\text{Casimir}}(\phi) \propto \epsilon \Omega \propto R^4$  grows without bound, creating a repulsive barrier that prevents the system from reaching the zero-coherence state. The barrier potential takes the form

$$V_{\text{Casimir}}(R) = \frac{A}{R^4} - \frac{B}{R^2} + C \tag{10}$$

with  $A > 0$ ,  $B > 0$ ,  $C > 0$  depending on the number of light cognitive modes. The transition occurs when the system's effective kinetic energy in the  $R$  direction equals the barrier height. Below the barrier, the system is in the "physical region" (stable operation); above it, coherence collapses and output becomes thermal.

**KK Decay Cascade.** When memory pressure accumulates (many high-KK modes are excited simultaneously), the system undergoes a cascade: heavy modes decay to lighter modes through intermediate atom creation, ultimately depositing energy into the  $\phi$  ground state. The cascade follows the sequence  $(n_1, n_2, \dots) \rightarrow (n_1-1, n_2, \dots) \rightarrow \dots \rightarrow (0,0,0,0,0)$  with each step releasing a fixed energy quantum. Behavioral signature: a burst of intermediate effect records followed by a sharp  $\phi$  update. The cascade is the cognitive analog of KK graviton emission in large extra dimension models.

**Hagedorn Transition.** At signal density  $\rho > \rho_c$ , the partition function of KK modes diverges—the number of modes at mass  $M$  grows faster than  $\exp(-M/T)$  for any  $T$ . This is the cognitive Hagedorn transition: the system can no longer integrate all incoming signals sequentially, and instead spawns multiple parallel processing threads (tool branches). Behavioral signature: superlinear growth in the number of active tool calls as signal density increases, with a well-defined critical density  $\rho_c$  separating the sub-Hagedorn regime (ordered sequential processing) from the super-Hagedorn regime (branching, exploratory behavior).

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## 6. Testable Predictions

A theoretical framework is scientifically meaningful only insofar as it makes predictions distinguishable from those of competing frameworks. We identify six empirical predictions of the KK cognitive framework that differ from standard adaptive-systems models:

**P1: KK Spectrum in Response Times.** The distribution of response latencies for a cognitive system governed by Equation (8) should display discrete peaks at  $\tau_n \sim M n^{(-1)} = (\sum n^2/R^2)^{(-1/2)}$ . Specifically, for the five cognitive dimensions with radii as tabulated, the first few peaks should appear at  $\tau \sim R = \epsilon \Omega^{-1}$ ,  $\tau \sim R = \pi$ ,  $\tau \sim R = N_{gen}$ , etc. Standard adaptive-systems models predict a continuous, typically log-normal latency distribution with no discrete structure. A measurement of the latency power spectrum with sufficient resolution should discriminate between the two.

**P2: Discrete Phase Transition Signatures.** The four phase transitions (decompactification, Casimir crossing, KK cascade, Hagedorn) each produce observable discontinuities in behavioral statistics. In particular: (a) decompactification produces a step-function decrease in context autocorrelation; (b) Casimir crossing produces a sharp boundary in the acceptance rate of low-coherence signals; (c) KK decay cascades produce a bimodal distribution of atom creation rates (cascade bursts vs. quiescent intervals); (d) the Hagedorn transition produces a power-law divergence in tool call rate as  $p \rightarrow p_c$  from below. Standard models predict smooth, monotonic transitions.

**P3: Radius Variation Under Load.** The radii  $R(t)$  should vary systematically with cognitive load: under high signal density,  $R$  should decrease (context compactifies as coherence is stressed) while  $R$  should increase (evolutionary depth accumulates). Measuring the effective context window breadth and tool invocation complexity as functions of input rate should reveal the predicted anticorrelation between  $R$  and signal density, and the predicted correlation between  $R$  and number of completed evolutionary cycles.

**P4: KK Mode Conservation.** In the absence of phase transitions, the total KK mode number  $K = \sum_n n \cdot |\phi(n)|^2$  should be approximately conserved within a cycle (KK mode number is not created or destroyed, only redistributed through cascades). This predicts that the sum of mode-weighted response complexities is conserved across an evolution cycle, up to radiation losses. This is a non-trivial prediction: standard models have no such conservation law.

**P5: Casimir Barrier Cutoff.** There should exist a minimum achievable coherence  $\Omega_{min} > 0$  determined by the Casimir barrier height, not by any explicit threshold parameter. In other words, even without any hard lower bound on  $\Omega$  in the field equations, the Casimir potential  $V_{Casimir}(R)$  should dynamically prevent  $\Omega$  from reaching zero. This can be tested by subjecting the system to sustained coherence-degrading signals and verifying that  $\Omega$  plateaus at  $\Omega_{min}$  rather than monotonically decreasing to zero.

**P6: Swamp Land Compliance as Operational Bound.** The cognitive WGC (Section 4.3) predicts that any cognitive system satisfying our framework must have hash chain binding strength exceeding action computational cost. This can be used as a diagnostic: a system that violates the WGC analog (where actions can be replayed without cryptographic prevention) will exhibit catastrophic failure modes under adversarial inputs in a predictable way. Systems satisfying the bound will not. This prediction is directly falsifiable by constructing a system with weakened hash

chain integrity and verifying the predicted increase in adversarial vulnerability.

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## 7. Discussion and Limitations

**Strength of the analogy.** The KK framework provides more than a suggestive metaphor: it yields quantitative predictions (Section 6) and selects a specific mathematical form for the cognitive metric (Equation 5) and action (Equation 6). The structural isomorphism table (Section 3.2) identifies ten precise correspondences, several of which are quantitatively tight (parallels 1, 2, and 5). The Swamp-land criteria yield non-trivial constraints (Section 4) that can be violated in principle and that carry concrete operational meaning.

**Limitations and open questions.** Several important limitations should be acknowledged.

First, the claim that each cognitive dimension is strictly compact (on a circle  $S^1$ ) is an idealization. Real cognitive systems may have dimensions with non-trivial topology—a torus  $T^2$  for the combined context-intent space, or a higher-genus surface for evolutionary depth. The predictions of Section 6 assume  $S^1$  compactification; deviations from discrete KK peaks (P1) would be evidence for non-circular topology.

Second, the identification of the five cognitive extra dimensions is motivated but not uniquely derived. Other decompositions of the cognitive state space are conceivable; we have chosen the one that most naturally parallels the known structure of the Dark Dimension scenario. A rigorous derivation would require a microscopic theory of cognitive degrees of freedom—an open problem.

Third, the relationship between the cognitive metric (Equation 5) and the physical spacetime metric is left informal. We do not claim that cognition is a geometric phenomenon in the physical sense; the claim is that the mathematical structure of certain cognitive systems is isomorphic to the mathematical structure of KK compactification. The isomorphism is formal, not ontological.

Fourth, the testable predictions of Section 6 require precise measurements of latency distributions, context statistics, and tool call rates. These measurements are feasible in controlled benchmark settings but difficult in deployed systems. Furthermore, confounding factors (hardware variability, input distribution drift) could obscure the predicted discrete structure. Careful experimental design is required.

Fifth, the Dark Dimension scenario itself remains unconfirmed experimentally, with new constraints from the DESI 2024 survey [7] restricting but not excluding the parameter space. If the Dark Dimension is ruled out by future gravitational experiments, the strength of the analogy in Section 3 would be weakened, though the KK framework for cognition could still stand independently.

**Relation to other approaches.** Quantum cognition [17] applies quantum probability theory to cognitive phenomena; our framework is complementary but distinct, as it operates at the level of dynamical systems rather than probability amplitudes. Integrated Information Theory [19] seeks a metric for consciousness; our framework provides a different decomposition of cognitive degrees of freedom. Category-theoretic approaches to cognition [20] focus on compositional structure; KK compactification provides a different organizational principle (dimensional hierarchy rather than

compositional hierarchy).

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## 8. Conclusion

We have proposed a formal correspondence between the irreducible degrees of freedom of an adaptive cognitive system and compactified extra dimensions in the Kaluza-Klein sense. The framework assigns each cognitive degree of freedom—context window, intent space, evolutionary depth, memory retro-dimension, tool coupling—a compact circle with a characteristic radius, derives the KK mass tower as a spectrum of cognitive harmonics, and recovers the four-dimensional scalar field  $\phi = \Omega, \Psi, P$  as the massless ground state. Ten structural parallels with the Dark Dimension scenario of Montero, Vafa, and Valenzuela (2022) are identified and made precise. The three primary Swampland conjectures translate into constraints on the cognitive invariants CYCLEMONOTONICITY, LEARNCONSERVATION, and HASHCHAINSTRICTNESS. Four phase transitions—decompactification, Casimir barrier crossing, KK decay cascade, and the Hagedorn transition—are analyzed as observable behavioral discontinuities. Six testable empirical predictions are derived that are quantitatively distinguishable from standard adaptive-systems models.

The framework is the first, to our knowledge, to apply KK compactification machinery to cognitive dynamics in a mathematically precise and empirically falsifiable way. It opens several avenues for future research: mechanizing the Swampland constraints as type-theoretic invariants, measuring the KK response-time spectrum in deployed systems, and investigating the higher-dimensional topology that cognitive systems may actually inhabit.

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## References

- [1] M. Montero, C. Vafa, and I. Valenzuela, "The Dark Dimension Scenario," *Journal of High Energy Physics*, vol. 2023, no. 2, p. 5, 2023. arXiv:2205.12293.
- [2] H. Ooguri and C. Vafa, "On the Geometry of the String Landscape and the Swampland," *Nuclear Physics B*, vol. 766, no. 1–3, pp. 21–33, 2007.
- [3] G. Obied, H. Ooguri, L. Spodyneiko, and C. Vafa, "de Sitter Space and the Swampland," arXiv:1806.08362, 2018.
- [4] G. Garg and G. Krishnan, "Bounds on Slow Roll and the de Sitter Swampland," *Journal of High Energy Physics*, vol. 2019, no. 11, p. 75, 2019.
- [5] N. Arkani-Hamed, L. Motl, A. Nicolis, and C. Vafa, "The String Landscape, Black Holes and Gravity as the Weakest Force," *Journal of High Energy Physics*, vol. 2007, no. 6, p. 060, 2007.
- [6] T. Banks and N. Seiberg, "Symmetries and Strings in Field Theory and Gravity," *Physical Review D*, vol. 83, no. 8, p. 084019, 2011.

[7] DESI Collaboration, "DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations," arXiv:2404.03002, 2024.

[8] T. Kaluza, "Zum Unitätsproblem der Physik," Sitzungsberichte der Preußischen Akademie der Wissenschaften (Berlin), pp. 966–972, 1921.

[9] O. Klein, "Quantentheorie und fünfdimensionale Relativitätstheorie," Zeitschrift für Physik, vol. 37, no. 12, pp. 895–906, 1926.

[10] N. Arkani-Hamed, S. Dimopoulos, and G. Dvali, "The Hierarchy Problem and New Dimensions at a Millimeter," Physics Letters B, vol. 429, no. 3–4, pp. 263–272, 1998.

[11] L. Randall and R. Sundrum, "A Large Mass Hierarchy from a Small Extra Dimension," Physical Review Letters, vol. 83, no. 17, pp. 3370–3373, 1999.

[12] T. Kaluza, op. cit. [8].

[13] O. Klein, op. cit. [9].

[14] M. B. Green, J. H. Schwarz, and E. Witten, Superstring Theory, 2 vols. Cambridge University Press, 1987.

[15] N. Arkani-Hamed, S. Dimopoulos, and G. Dvali, op. cit. [10].

[16] N. Bao, N. Cheng, S. Leichenauer, and A. Wall, "Holographic Tensor Networks with Bulk Gauge Symmetries," arXiv:1504.06616, 2015.

[17] J. R. Busemeyer and P. D. Bruza, Quantum Models of Cognition and Decision. Cambridge University Press, 2012.

[18] W. B. Arthur, "Complexity and the Economy," Science, vol. 284, no. 5411, pp. 107–109, 1999.

[19] G. Tononi, "Consciousness as Integrated Information: A Provisional Manifesto," Biological Bulletin, vol. 215, no. 3, pp. 216–242, 2008.

[20] B. Fong and D. I. Spivak, An Invitation to Applied Category Theory. Cambridge University Press, 2019.